

Does oil price volatility delay natural gas infrastructure investment? Evidence from 1997-2025*

Sofia P. Baker[†]

Abstract

Price volatility delays irreversible investments like natural gas infrastructure. A possible contributor to such volatility is spillover from crude oil to natural gas markets. Previous studies have shown a declining trend in correlation of oil and natural gas volatilities. However, evidence is not yet available for how volatility spillovers have responded to recent market innovations. Therefore, the goal of this study was to analyze how volatility spillovers among US oil and gas spot prices change through 1997-2025. To this end, I use the Diebold-Yilmaz index on realized volatilities, the Bai-Perron structural break test on total spillovers, and DCC-GARCH for modeling correlation of returns. My results show volatility spillovers have reduced from 1997-2025, with the lowest spillover occurring after 2019. Moreover, DCC-GARCH shows correlation among returns declines at the end of the sample. Finally, directional results indicate both oil-to-gas and gas-to-oil spillovers are weakened with neither market being the dominant volatility transmitter. Together, these findings support increased independence of oil and gas market volatilities in a longer timeframe than formerly observed and suggest that oil market volatility is not a significant contributor to the natural gas infrastructure investment decision.

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[†]Department of Economics, University of Missouri

1 Introduction

While crude oil is easy to work with, natural gas depends on pipelines, specialized storage facilities and terminals for conversion to liquefied natural gas (LNG) for overseas trading (Liang et al., 2012) because it is a gas. Lack of such infrastructure has limited switching from crude oil to natural gas, leaving open an arbitrage where natural gas is much cheaper than oil in terms of underlying energy content (Liang et al., 2012; Brown and Yücel, 2008). At the same time, US citizens and policymakers are beginning to recognize that increasing energy supply is critical for supporting standards of living and to compete in the race for AI-driven industrial growth (Benich et al., 2026; Kay et al., 2026). To meet this energy need, it is important for the US to leverage its advantage as the highest producer of natural gas in the world (IEA, 2026), which involves construction of natural gas infrastructure in the near future. Despite the benefits to expanding the natural gas infrastructure network, volatile prices have quenched incentives for investment in transport networks because builders need to be certain the price will stay high enough to cover large upfront costs.

In particular, traditional firm investment theory suggests it is rational to invest if expected value is marginally higher than expected cost, but the real options literature shows this is not profitable in the case of sunk cost investments (Kellogg, 2014). Specifically, the real options literature shows the value of waiting to invest increases with expected price volatility holding expected return constant. Kellogg (2014) provides empirical evidence supporting this theory using firm-level data from the Texas oil drilling market. In this case, firms have reserved rights to a well but can wait to do expensive drilling and choose to wait longer when expected price volatility is high. Décaire et al. (2020) confirm Kellogg’s findings within the natural gas drilling market. Likewise, natural gas infrastructure investments require large sunk costs (Von Hirschhausen and Neumann, 2008) and are therefore delayed by price volatility.

While applicable for all sunk cost investments, the price volatility disincentive is particularly important for the natural gas industry because gas is among the most volatile commodities in the US economy. This volatility stems from multiple factors, including the

seasonal, largely unpredictable, nature of demand and the difficulty to store natural gas (Liang et al., 2012). Natural gas in the US is mainly used for electricity and direct heating, whose consumption fluctuates wildly with season (Mu, 2007). Moreover, weather specifics are hard to forecast and regularly shock the market leading to price volatility (Mu, 2007). Regarding storage, gas is usually held underground with pipeline networks to add and withdraw gas (Rubaszek and Uddin, 2020), which contrasts with crude oil that is easily kept in barrels. Natural gas production is virtually constant across the year (Liang et al., 2012), so producers do a balancing act between how much they produce and how much they are able to store. If they produce beyond their existing storage capacity, they are forced to sell on the spot market and crash the price because short-run demand is inelastic (Nick and Thoenes, 2014; Gautam and Paudel, 2018). Instead of being dictated by global storage and demand as oil is, natural gas is mainly bounded local storage capacity and weather (Mu, 2007; Liang et al., 2012) which creates a very volatile market.

Beyond natural gas fundamentals, oil-to-gas volatility spillover may also be contributing to natural gas volatility (Ebrahim et al., 2014; Gong et al., 2021). Volatility spillovers occur when volatility in one asset creates volatility in another (Diebold and Yilmaz, 2012), which introduces uncertainty in the receiving market. Although oil is not as volatile as natural gas, it could have a significant effect on gas volatility if gas supply and demand depend on oil dynamics. For example, treasury rates are less volatile than stock prices, but significant volatility spillover is still found from treasuries to stocks (González-Urteaga et al., 2022). Interest rates are a primary determinant of corporate profitability and therefore uncertainty in the less volatile market influences uncertainty in the more volatile market. Similarly, a small amount of volatility in oil could create volatility in natural gas because of the coproduction and substitution relationships of the two products.

Regarding coproduction, oil and gas are often extracted together through traditional wells, or more recently, hydraulic fracturing (i.e., fracking). Fracking was developed in the shale revolution where existing technologies were combined in new ways to recover oil and

gas from previously unusable areas (Golden and Wiseman, 2014). The shale revolution began in the early 2000's and gained momentum around 2009 when US production began to show large increases (Sieminski, 2014). The first movers in the shale revolution were small firms (Golden and Wiseman, 2014), many of whom struggled during the Great Financial Crisis (GFC) and became consolidated at this time (Livingston, 2014). Coming out of the GFC, government stimulus led remaining businesses to flood the market with oil and gas (Al Suwailem and Selemankhel, 2021) which created a divergence between oil and gas spot prices because drillers treated natural gas as a byproduct of high-value oil production (Stern, 2020). The focus on oil over gas left producers willing to sell massive quantities of associated gas (i.e., gas which is extracted as a coproduct of oil) at any price, with total gas production doubling from 2005-2025 (EIA, 2026).

Beyond shared production, oil and gas markets may be tied because natural gas is often considered a substitute for refined petroleum products, with natural gas having advantages over petroleum through minimal refining, much cleaner burning, and relatively low prices (Liang et al., 2012). Historically, oil and gas have been imperfect substitutes with natural gas mainly used for direct heating through combustion and creating electricity (Liang et al., 2012) and petroleum typically refined into end products such as gasoline, diesel and heating oil (Jin and Xu, 2024; Kilian, 2016). While it is possible to use petroleum for electricity generation, burning gas is much more efficient (Oh et al., 2021), making natural gas dominant in this space (EIA, 2024c) but with many plants carrying petroleum capacity to shield against gas price spikes (Gal et al., 2019). Moreover, natural gas competes with petroleum derivatives for residential and industrial heating and steam generation (Liang et al., 2012; Einstein et al., 2001). Both gas and petroleum are also inputs for plastics and petrochemicals, where the two energy sources are substitutable in the long run (Kapsalyamova and Paltsev, 2020). Similarly, in transportation, petroleum derivatives (e.g., diesel, gasoline, kerosene) have been dominant (EIA, 2019b), but technologies to switch from using oil to gas engines have existed for a long time (Painuly and Parikh, 1993) indicating a potential for increased switching in

the long run.

Given the economic connection of the two markets, studying oil-to-gas volatility spillover is necessary to understand what factors firms are considering in the decision to invest in natural gas infrastructure. If oil-to-gas spillover is limited, then conflicts like the US-Iran war do not necessarily delay natural gas infrastructure development. Previous studies of volatility spillovers among oil and gas surrounding the shale revolution give mixed evidence regarding a disruption in market behavior. For the period covering the early 2000's, studies find bidirectional spillovers (Ewing et al., 2002; Lin and Li, 2015; Lovcha and Perez-Laborda, 2020), thereby indicating that the two markets were connected. After the shale revolution, studies generally find weaker spillovers (Lovcha and Perez-Laborda, 2020; Zhu et al., 2018; Gong et al., 2021) but differ on how much spillovers were reduced. For instance, Zhu et al. (2018) find near-zero spillovers with weak evidence of oil-to-gas spillover and no evidence of the reverse. In contrast, Gong et al. (2021) and Lovcha and Perez-Laborda (2020) study the periods of 2005-2019 and 1994-2018, respectively, and find evidence of some remaining spillover in periods following the shale revolution, but disagree on the dominance of oil versus gas, with Gong et al. (2021) finding oil as the dominant transmitter of volatility while Lovcha and Perez-Laborda (2020) find alternating dominance.

More recently, Kočenda and Moravcová (2024) finds low to moderate connectedness of volatility spillover across 1997-2023, with some elevation surrounding the Russia-Ukraine war. Regarding methods, early studies often adopted GARCH-BEKK (Ewing et al., 2002; Lin and Li, 2015) which has a clear interpretation of results for volatility spillover questions but is often computationally infeasible because of how many parameters are needed. For this reason, the GARCH literature has largely shifted towards DCC-GARCH (Efimova and Serletis, 2014) which reduces dimensionality and efficiently captures time-varying correlation of returns. However, in simplifying the model, the capacity to estimate volatility spillovers is lost. Therefore, the Diebold-Yilmaz connectedness index has emerged as a parsimonious way to analyze volatility spillovers between oil and gas (Lovcha and Perez-Laborda, 2020;

Gong et al., 2021; Kočenda and Moravcová, 2024) and is based on estimating a VAR with realized volatilities of both assets and conducting a Generalized Forecast Error Variance Decomposition (GFEVD) (Diebold and Yilmaz, 2012).

I contribute to the above literature by showing the time series of volatility spillover among oil and natural gas alone across an extended time span. Previous studies (Gong et al., 2021; Kočenda and Moravcová, 2024) which add more energy commodities to the Diebold-Yilmaz system have the benefit of showing connectedness within broad sectors (e.g., energy commodities), but have a tradeoff of blurring oil-gas dynamics. In contrast, Lovcha and Perez-Laborda (2020) applies Diebold-Yilmaz analysis to oil and gas futures alone, thereby giving a clear picture of spillover among these markets using daily total, directional and net spillover indices from 1994-2018. Most recently, Kočenda and Moravcová (2024) gives oil-to-gas and gas-to-oil spillover but does not show a time varying index, instead providing estimates in aggregated windows from 1997 to February 2023. Such aggregation often oversimplifies dynamics because volatility spillover tends to occur in clusters rather than in steady amounts across a window (Lovcha and Perez-Laborda, 2020; Diebold and Yilmaz, 2012). Therefore, I extend the time frame under consideration for daily volatility spillovers among oil and natural gas spot prices for the US.

This is significant because of fundamental market changes which have been occurring in recent years, particularly a need for greater energy supplies. Specifically, the electricity reliability council of Texas (ERCOT) has reported they are expanding grid capacity at record speed to keep up with data center demand (EIA, 2025). Similar data center openings are happening all over the country, with estimates indicating 4% of US electricity consumption came from data centers in 2023 (Offutt and Zhu, 2025). To this end, data centers are connecting to local gas-powered grids at a breakneck pace, and some are even proposing to pipe in natural gas directly to generate power on-site (Bickel and Carr Smyth, 2026). Beyond domestic usage, more energy is needed outside the US due to geopolitical instability. For instance, LNG exports to Europe have increased following the Russia-Ukraine war and

sabotage of the Nordstream natural gas pipeline running from Russia to Western Europe (EIA, 2025).

Therefore, the goal of this study was to analyze how volatility spillovers among WTI and Henry Hub spot prices change through 1997-2025. To this end, I estimate the Diebold-Yilmaz connectedness index (Diebold and Yilmaz, 2012) in one year rolling windows on realized volatilities of oil and gas prices to obtain total and directional spillover measures. Next, to determine if volatility spillover changes significantly across time, I use the Bai-Perron structural break test (Bai and Perron, 1998) on the total spillover index. Finally, given the importance of time-varying volatility in crude oil and natural gas markets, I test previous findings of reduced correlation of returns among oil and gas (Dahl et al., 2012; Hasanli, 2024; Wang et al., 2025) using a DCC-GARCH model to control for effects of time-varying volatility. My results indicate that total volatility spillovers have reduced since 1997, with the lowest spillover regime appearing after 2019. Moreover, directional results indicate both oil-to-gas and gas-to-oil spillovers are weakened across the sample period. Lastly, DCC-GARCH results show time-varying positive correlation of returns with near-zero correlation at the end of the sample. These findings are novel because they show independence of oil and gas market volatilities in a longer timeframe than previously observed. This gives insight into the factors firms are considering in the expansion of gas infrastructure and suggest that effects of an increasingly uncertain crude oil market on the gas infrastructure space are limited.

2 Data

Regarding data, I use the West Texas Intermediate (WTI) crude oil spot¹ and Henry Hub natural gas spot prices,² both from the Energy Information Administration. The units of WTI spot are US dollars per barrel and Henry Hub spot are US dollars per million British

¹Downloaded on 2025-12-28 from <https://www.eia.gov/dnav/pet/hist/RWTCD.htm>

²Downloaded on 2025-12-18 from <https://www.eia.gov/dnav/ng/hist/rngwhhdD.htm>

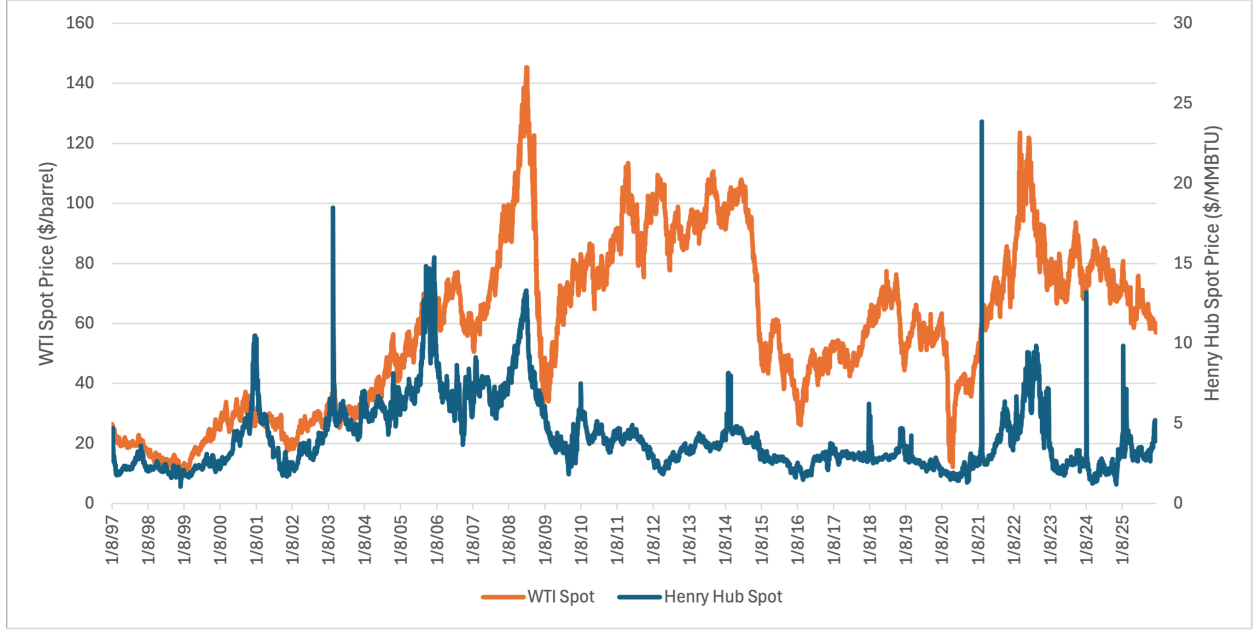


Figure 1: WTI Crude Oil and Natural Gas Spot Prices.

thermal units (MMBTU). To clean these data, I keep dates where both price series are available, leaving a sample period of trading days from January 1997 – December 2025, shown in Figure 1 where before 2009 the two prices moved together but subsequently diverged, and towards the end of the sample, begin to converge.³ Analyses use log returns and realized volatilities which are calculated as the standard deviation of the log returns in 20-day rolling windows (Figure 2). To investigate the stationarity of the crude oil and natural gas returns and volatilities, I use the Phillips-Perron unit root test (Phillips and Perron, 1988), where the null hypothesis is that the time series has a unit root. I reject the null hypothesis of a unit root for both returns and volatilities.

³On April 20th, 2020, the WTI spot price went negative for the first time in history due to rapid demand decreases from the COVID pandemic and returned to positive values the following day (Bourghelle et al., 2021). The April 20th observation has been removed from Figure 1 and log returns from this observation were not calculated for April 20th and April 21st.

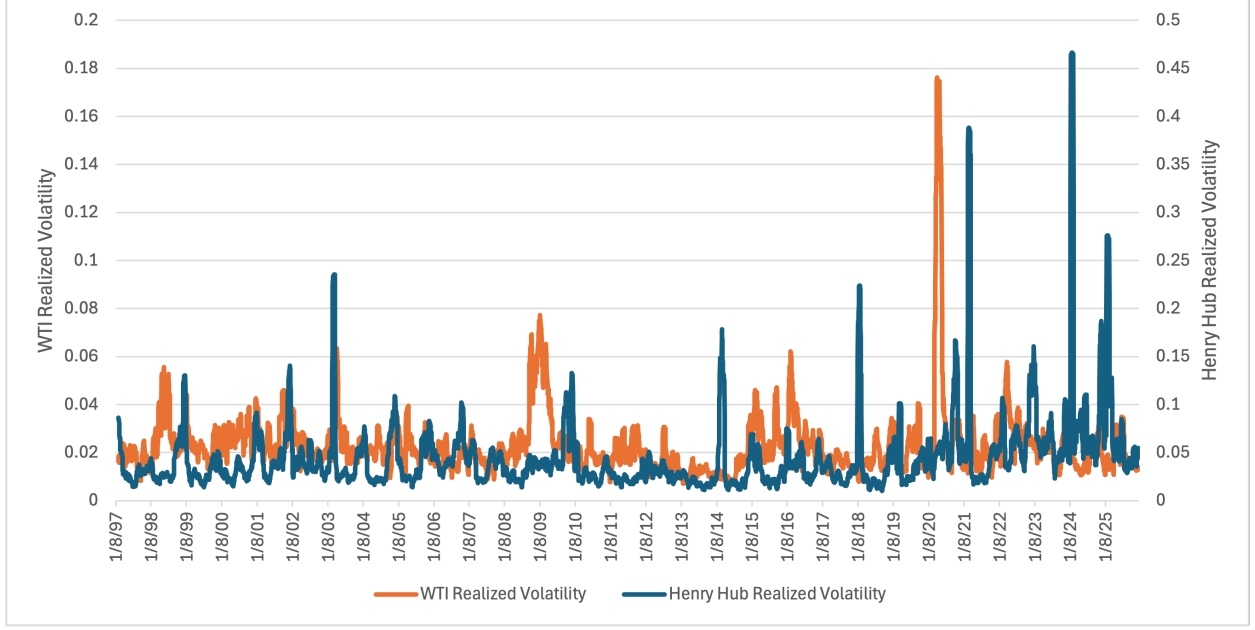


Figure 2: WTI Crude Oil and Natural Gas Realized Volatility in 20-day rolling windows.

3 Methods

3.1. Diebold-Yilmaz Spillover Index

The primary method used is the Diebold-Yilmaz spillover index (Diebold and Yilmaz, 2012) which is applied to volatility estimates to understand their overall connectedness and directionality of effects. To this end, I calculate the Diebold-Yilmaz spillover index in four directions:

1. Spillovers from natural gas to oil
2. Spillovers from oil to natural gas
3. Total spillover (1+2)
4. Net spillover (1-2)

In particular, consider a covariance stationary N -variable $\text{VAR}(p)$

$$x_t = \sum_{i=1}^p \Phi_i x_{t-i} + \varepsilon_t \quad (1)$$

where x_t is a vector of realized volatilities and ε_t is a vector of iid disturbances. The moving average representation is

$$x_t = \sum_{i=0}^{\infty} A_i \varepsilon_{t-i} \quad (2)$$

where the $N \times N$ coefficient matrices A_i obey the recursion

$$A_i = \Phi_1 A_{i-1} + \Phi_2 A_{i-2} + \dots + \Phi_p A_{i-p} \quad (3)$$

with A_0 being an $N \times N$ identity matrix and $A_i = 0$ for $i < 0$. The spillover indices are based on an H -step ahead GFEVD given by

$$\theta_{ij}^g(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} (e_i' A_h \Sigma A_h' e_i)} \quad (4)$$

where a GFEVD is used instead of a traditional FEVD to make the results invariant to variable ordering in the VAR. Moreover, σ_{jj} is the standard deviation of the error term for the j -th equation, e_i is a selection vector with one as the i -th element and zero otherwise, Σ is the variance matrix for error vector ε and A_h is the $N \times N$ moving average coefficient matrix from the VAR of realized volatilities for the h -th lag. In a GFEVD, the row sum $\sum_{j=1}^N \theta_{ij}^g(H)$ does not equal one as it would in a traditional FEVD and therefore the decompositions are normalized as

$$\tilde{\theta}_{ij}^g(H) = \frac{\theta_{ij}^g(H)}{\sum_{j=1}^N \theta_{ij}^g(H)} \quad (5)$$

which makes $\sum_{j=1}^N \tilde{\theta}_{ij}^g(H) = 1$ and $\sum_{i,j=1}^N \tilde{\theta}_{ij}^g(H) = N$. Equation (5) shows the percentage of forecast error variance (FEV) of variable i explained by shocks to variable j . This formula is rewritten as

$$S_{.i}^g = \frac{100 \sum_{j=1, j \neq i}^N \tilde{\theta}_{ji}^g(H)}{\sum_{i,j=1}^N \tilde{\theta}_{ji}^g(H)} = \frac{100 \sum_{j=1, j \neq i}^N \tilde{\theta}_{ji}^g(H)}{N} \quad (6)$$

to show spillovers from market i to all other markets j as a fraction of total FEV (i.e., FEV of oil and gas markets combined which equals N from normalization above). Given

the rescaling from equation (6), the directional spillover measures are bounded from zero to $100/N$ while total spillovers are bounded from zero to 100. Both directional and total spillovers show the magnitude of volatility connectedness relative to total FEV while net spillovers indicate differences in strengths of directional indices. Therefore, the units for net spillover change from percent of total FEV to percentage points (pp) which are bounded from -50 to 50. Both oil-to-gas and gas-to-oil spillovers are calculated across the entire sample in 252-day rolling windows (equivalent to one year in trading days) and are combined to produce total and net spillovers. Finally, $p = 4$ and $H = 10$ are chosen to match Diebold and Yilmaz (2012).

3.2. Bai-Perron Structural Break Test

To test for changes in the connectedness of oil and natural gas volatilities I run the Bai-Perron structural break test (Bai and Perron, 1998) on a regression of the total spillover index from the Diebold-Yilmaz analysis on an intercept. Specifically, the purpose of this step is to determine whether there exist statistically significant differences in the level of volatility connectedness across the sample. In particular, consider a linear regression with m breaks ($m + 1$ regimes) of total volatility spillover y_t onto a an intercept δ_k given by

$$y_t = \delta_k + u_t \quad (t = T_{k-1} + 1, \dots, T_k) \quad (7)$$

where δ_k is regime-dependent with a separate value for each $k = 1, \dots, m + 1$. Indices $(T_1, \dots, T_m)$ are the unknown break points where I use convention $T_0 = 0$ and $T_{m+1} = T$. To find the optimal breaks for a given m , all possible times of breaks $(T_1, \dots, T_m)$ are run and selected by minimizing the sum of squared residuals. This is done for several break numbers m and the BIC is used to find the optimal number of breaks.

3.3 DCC-GARCH

To confirm previous findings of reduced connection of price levels in oil and natural gas mar-

kets, I estimate a Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity (DCC-GARCH) model (Engle, 2002). The conditional covariance matrix of returns is defined as

$$E_{t-1}(r_t r_t') \equiv W_t \quad (8)$$

and is decomposed into the product given by

$$W_t = D_t R_t D_t \quad (9)$$

$$D_t = \text{diag}\{\sqrt{w_{i,t}}\}, \quad (10)$$

$$w_{i,t} = E_{t-1}(r_{i,t}^2) \quad (11)$$

where D_t is a standard deviation matrix and R_t is the conditional correlation matrix. In particular, because the vector of oil and gas log returns r_t is 2×1 and the correlation of a variable with itself is always one, R_t can be rewritten as

$$R_t = \begin{bmatrix} 1 & \rho_t \\ \rho_t & 1 \end{bmatrix} \quad (12)$$

where ρ_t is the dynamic conditional correlation of oil and gas standardized returns which I aim to estimate with DCC-GARCH. In addition to capturing a time-varying correlation of returns, each variable (oil log returns and natural gas log returns) individually follows a GARCH(1,1) specification which models current conditional variance as a function of last period's error and conditional variance. By fitting a GARCH(1,1) to both series, the DCC-GARCH is able to standardize the returns to control for time-varying volatility in both assets. In particular, the DCC-GARCH is formulated by

$$r_t | \mathcal{F}_{t-1} \sim N(0, D_t R_t D_t) \quad (13)$$

$$D_t^2 = \text{diag}\{\omega_i\} + \text{diag}\{\alpha_i\} \circ r_{t-1}r'_{t-1} + \text{diag}\{\beta_i\} \circ D_{t-1}^2 \quad (14)$$

$$\eta_t = D_t^{-1}r_t \quad (15)$$

$$Q_t = \Omega(1 - a - b) + a\eta_{t-1}\eta'_{t-1} + bQ_{t-1} \quad (16)$$

$$R_t = \text{diag}\{Q_t\}^{-1}Q_t\text{diag}\{Q_t\}^{-1} \quad (17)$$

where $\circ$ denotes the Hadamard product of two identically sized matrices which is computed as element-by-element multiplication. The first equation assumes normality of returns conditional on past information to justify Maximum Likelihood Estimation (MLE) and the second equation shows each asset as following a univariate GARCH model. Moreover, ω_i is the unconditional variance for asset i , α_i is the ARCH coefficient for asset i , β_i is the GARCH coefficient for asset i , η_t is the vector of standardized returns which is mean zero and variance one for each series, Q_t is the conditional covariance of standardized returns, Ω is the unconditional covariance of standardized returns, a indicates how fast the correlation of returns responds to current shocks and b shows persistence of correlation. The DCC-GARCH model is estimated using two-stage Quasi-Maximum Likelihood Estimation (QMLE) where the GARCH(1,1) equations are estimated first, and the resulting coefficients are used to obtain the dynamic conditional correlation.

4 Results

Table 1: Summary of volatility spillover results within regimes identified by Bai-Perron test.

Regime	Start Date	End Date	Duration	Average Total Spillover (%)	Average Net Spillover (pp)
1	Feb 1998	Apr 2002	4.2 years	2.45	0.24
2	Apr 2002	Jun 2006	4.2 years	4.11	-0.54
3	Jun 2006	Jan 2015	8.4 years	2.16	0.58
4	Jan 2015	Mar 2019	4.2 years	4.16	0.47
5	Mar 2019	Dec 2025	6.8 years	1.69	0.24

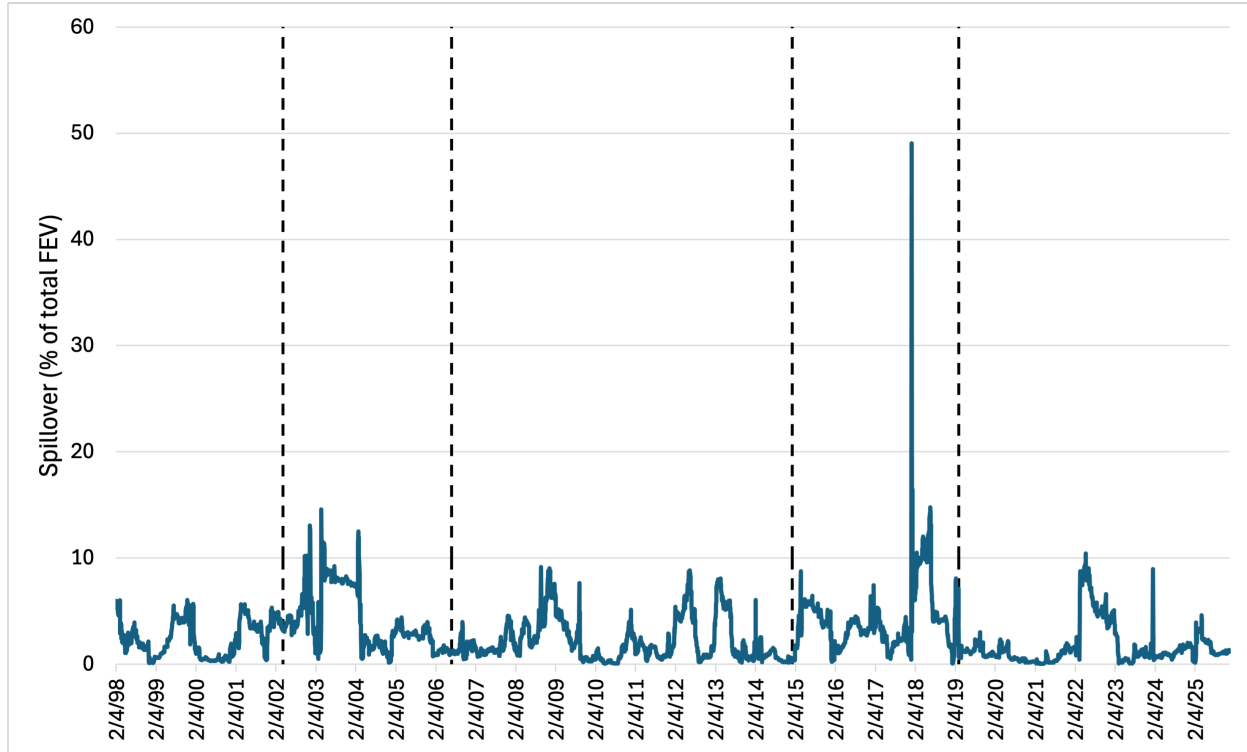


Figure 3: Diebold-Yilmaz total volatility spillover index. Vertical lines indicate structural breaks found by Bai-Perron test on 2002-04-10, 2006-06-28, 2015-01-05 and 2019-03-07.

4.1. Diebold-Yilmaz Spillover Index

Figure 3 shows the total volatility spillover index where context for interpreting the units (percentage of total forecast error variance) comes from past applications of the Diebold-Yilmaz index in other markets. For instance, Diebold and Yilmaz (2012) study spillovers between US stock, bond, currency and commodity prices and find that total spillovers range from 10-30%, on average being 12%, with highs of $\sim 30\%$ reached during the GFC when there was significant transmission of risk information across markets. Additionally, Gong et al. (2021) use the Diebold-Yilmaz index to analyze oil and natural gas volatilities together with heating oil and gasoline, which brings the total spillover range to 35-60% indicating high connectedness. Moreover, the 49% sample high total spillover during January 2018 in Figure 3 is comparable in magnitude with the sample high of Lovcha and Perez-Laborda (2020) at 45% during February 2015, which the authors interpret as high connectedness. While the magnitudes are similar, the timing of the sample high differs, which may be

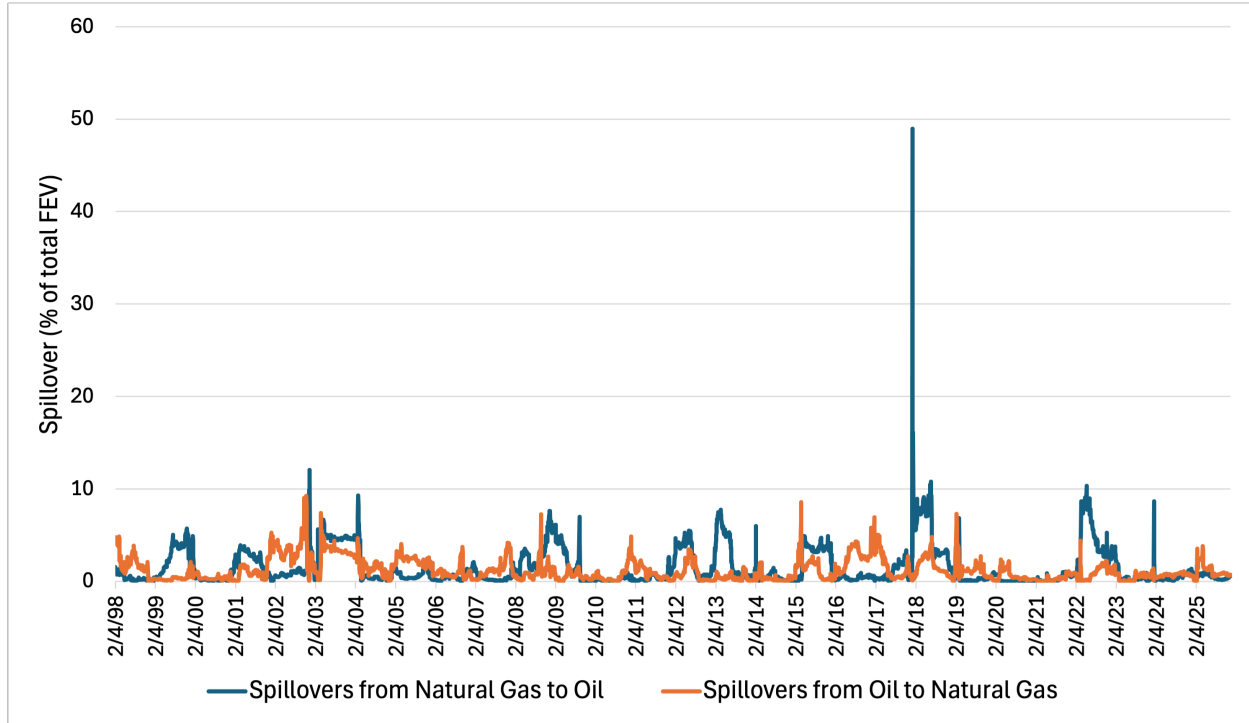


Figure 4: Diebold-Yilmaz directional volatility spillover indices.

attributed to a distinction between futures prices used by Lovcha and Perez-Laborda (2020) and spot prices used here for the purpose of analyzing supply dynamics with less influence of financial factors such as interest rates (Cheng and Xiong, 2014; Pindyck, 2001). Given this scale, 5-10% total spillovers indicate mild connectedness between oil and gas volatilities, with above 10% total spillover being moderate. Overall, Figure 3 indicates mild to moderate connectedness early, brief high connectedness in early 2018, and reduced spillovers at the end of the sample, indicating that crude oil and natural gas are not significant contributors for the other markets' volatility.

Specifically, total volatility spillovers show regular variation ranging from near zero spillovers (e.g., 2000, 2006) to 5-10% spillovers (e.g., 2003). The total spillover index contains periods beyond 10% in October 2002-March 2004, January-June 2018, and May 2022. Figure 3 also includes vertical lines which separate regimes of total spillover based on Bai-Perron test results. The early sample period before the structural break in April 2002 shows mild spillover, while the second regime contains periods of moderate spillover. Specifically, Octo-

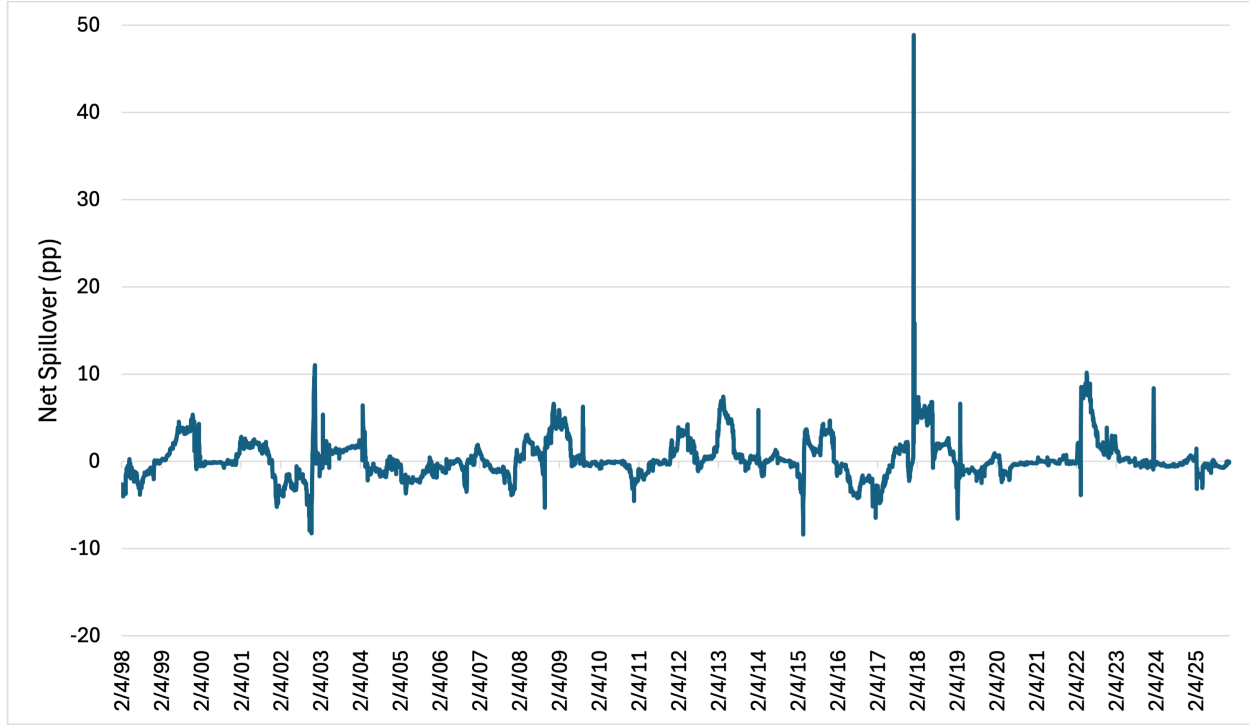


Figure 5: Diebold-Yilmaz net volatility spillover index. Positive values indicate gas-to-oil spillover is stronger than oil-to-gas spillover.

ber 2002-March 2004 was marked by relatively high gas-to-oil and oil-to-gas spillovers both (Figure 4), creating a compounding effect leading to total spillovers beyond 10% at various points across the year and a half period. At this time, the Afghanistan and Iraq wars were escalating (Sollinger et al., 2008), thereby increasing oil-to-gas spillovers from uncertainty in Middle Eastern oil supplies reaching the rest of the world, especially given that Iraq was an oil producer (Bamber et al., 2023). Moreover, the US was highly reliant on petroleum imports at this time, with imports peaking in 2005 (EIA, 2023a), which plausibly increases the impact of oil volatility on other domestic markets (i.e., natural gas). Additionally, gas-to-oil spillovers could be explained by uncertainty in the oil market being compounded by natural gas reaching record highs of \$18 per MMBTU in February 2003 surrounding harsh winter weather at a time of low inventories (FERC, 2003).

After this time, spillovers return to a range from 0-10% indicating mild connectedness of volatility. The third regime is by far the longest regime identified by the Bai-Perron

Table 2: DCC-GARCH coefficients.

Parameter	Value	Std. Error
ω_n	0.29	0.056
α_n	0.14	0.011
β_n	0.86	0.011
$\alpha_n + \beta_n$	0.99	
ω_o	0.10	0.027
α_o	0.07	0.010
β_o	0.91	0.013
$\alpha_o + \beta_o$	0.98	
a	0.001	0.001
b	0.996	0.003
$a + b$	0.997	

test extending from June 2006 - January 2015. Total spillover remains under 10% and on average Regime 3 contains lower connectedness than earlier regimes, consistent with previous studies finding reduced spillover in the post-shale revolution period (Lovcha and Perez-Laborda, 2020). The shale revolution took off $\sim$ 2009, but began in the early 2000's (Sieminski, 2014), with previous studies emphasizing that the transition between pre-shale and post-shale regimes occurred slowly (e.g., Hasanli (2024) on the oil-gas cointegrating regression). I support the finding of a relative decoupling in volatilities after the shale revolution because average total spillover (Table 1) declined following 2006 when the shale revolution had initiated. This period of relatively low spillover extends to January 2015 which follows an OPEC decision in November 2014 to maintain high production while facing low prices to drive out higher cost US shale producers (Behar and Ritz, 2016).

Therefore, Regime 4 contains the global oil supply glut in 2015 where I observe relatively high oil-to-gas spillover (Figure 4), which could be explained by traders assuming low oil prices would cause natural gas supply declines. Next, total spillovers reach a sample high of 49.1% on 2018-1-2, which appears and dissipates quickly (previous observation had 1.7% spillover and next observation had 3.1% spillover). Outside of this peak, the next highest value is 16.5% on 2018-1-8, and both large spillovers can be attributed to gas-to-oil spillovers (49.0% on 2018-1-2 and 16.1% on 2018-1-8 in Figure 4). The natural gas price was very

volatile at this time, with the price increasing from \$3.69 to \$6.24 on 2018-1-2 and dropping to \$2.89 by 2018-1-8. This period of volatility was caused by the bomb cyclone event Winter Storm Grayson in early 2018 which caused heavy snowfall in the US Northeast and was characterized by rapid acceleration of storm effects, hence the bomb cyclone name (Brick, 2018; Yum, 2022; LeComte, 2019). Moreover, this period is characterized by very low oil market volatility, ranging from 0.013 to 0.009 compared to a sample average of 0.023, so any change in volatility appears amplified. Regarding later periods surrounding January to June 2018, total volatility spillover remains above 10% for nearly the entire period largely due to gas-to-oil spillovers which stay near 8%. The oil price was generally not volatile from January-June 2018 but what little volatility did appear could be attributed to the natural gas price given that the forces creating uncertainty in natural gas (i.e., the winter storm) created uncertainty for the demand of petroleum derivatives, notably heating oil in the northeast which was heavily affected by the storm (EIA, 2024b) and temporary switching to diesel generators during gas shortages (Gal et al., 2019).

The final structural break in Figure 3 occurs on 2019-03-07. After this time, the connection of the two markets appears to have reduced because the periods of low volatility spillover extend. Specifically, there is a notable period of low spillover from April 2019 - February 2022 where total spillover rarely went above 2%. Total spillover remained under 2% in an earlier period (February 2000 – March 2001) but for a much shorter time. The reduction in volatility spillovers late in the sample, and thus the 2019 structural break, could be explained by increased differentiation among natural gas and petroleum uses. The share of natural gas used for electricity where petroleum is difficult to substitute has increased, suggesting that the substitution relationship has weakened across time. Supporting this, the fraction of natural gas consumed by the electric power sector in the US (Appendix Figure A7) has increased from 18% to 39% during 1997-2025 as natural gas plants have replaced coal plants (Teller, 2016) and more electricity has been generated overall (EIA, 2024a). Given this, it is plausible that demand shocks in either market used to be more impactful in the

other market because oil and gas were used for similar tasks. Moreover, since natural gas demand for power is increasing, this would suggest that pure gas wells are becoming more valuable which also reduces the impact that oil prices have on natural gas supply decisions.

Highlighting the effect of differentiated uses are two significant price changes from April 2019 - February 2022 surrounding demand shocks in petroleum during COVID (Bourghelle et al., 2021) and natural gas during Winter Storm Uri (ENVERUS, 2021). At this time, a large fraction of gas went to residential use and electric power (Appendix Figure A7), where crude oil demand was mostly used for transportation and industrial applications (EIA, 2023b). Turning to the spot prices in Figure 1, a drawdown in the petroleum price in April 2020 following the onset of the pandemic occurs. This can be explained by reduced industrial activity across the globe, which affected the US petroleum market through reduced demand from both US and international industry because the petroleum market was globally integrated (Hasanli, 2024). In April 2020, natural gas prices were already low and retained their level because during the pandemic, individuals within the US still needed to power and heat their homes and decreases in natural gas demand from industrial production were offset by increases in home usage during quarantines (Cieřlik et al., 2022). Similarly, volatility in Henry Hub in February 2021 from Winter Storm Uri did not appear in the petroleum market because during the storm, while supply of both products was reduced from freeze-offs of wells (ENVERUS, 2021), demand for the two products responded in very different ways. For natural gas, heating and electricity demand skyrocketed from the cold weather (ENVERUS, 2021), while petroleum demand decreased from temporary closures of refineries in the South (Melaku et al., 2023) and reduced use of transportation fuels.

Together, these periods of high volatility from April 2019 – February 2022 did not translate to elevated volatility in the other market from the separate use cases of the products. However, this period of low spillover is interrupted by a period of higher volatility spillovers around May 2022. The Russia-Ukraine conflict at this time led to long-term increases in demand and volatility of the US gas market as uncertainty was brought into the domestic

market based on global geopolitical risk. Since the problem originated from the natural gas market, particularly Russian gas no longer entering Europe (Di Bella et al., 2024), the May 2022 spike in total spillover can also be attributed to large gas-to-oil spillover (Figure 4). Another period of low spillovers appears from February 2023 - December 2025, with the exception of a peak in January 2024 which was a large gas-to-oil spillover from another winter storm.

Turning to directional spillovers in Figure 4, both oil-to-gas and gas-to-oil spillovers show extended periods of low spillover at the end of the sample. Regarding magnitudes, equation (6) bounds the directional measures from zero to $100/N = 50$ while total spillover in Figure 3 is bounded from zero to 100. Given the rescaling, 5-10% directional spillovers now indicate moderate connectedness, with above 10% spillover indicating high connectedness. The early periods of the oil-to-gas spillover index show moderate spillovers, for instance appearing at highs of 9% in November 2002 and 7% in March 2003 during wars in Afghanistan and Iraq. Beyond these peaks, oil-to-gas spillovers show a declining trend from 2003 - 2007 and highs of 7% in September 2008 during the GFC and 8% in March 2015 in the oil supply glut. Later in the sample the typical range appears to be lower, and after the structural break in 2019, the peaks become smaller and occur less often. Specifically, spillover remains from 0-2% with the exception of 4% values in March 2022 and April 2025.

The trend in gas-to-oil spillovers shows periods of moderate to high spillover early, a peak in 2018 indicating high connectedness, and extended periods of low spillover post-2019 with several remaining peaks (i.e., Russia-Ukraine war and January 2024 winter storm). Gas-to-oil spillovers first go above 5% in November 1999 and appear at local highs of 12% and 9% in December 2002 and February 2004, respectively. Between March 2004 - December 2007 spillover is under 2%, rises again through 2008 – August 2009, and declines almost to zero from September 2009 - November 2011. Next, spillover rises to the 4-8% range until January 2016 – September 2017 when it is under 2% again. Next, during Winter Storm Grayson in January 2018 gas-to-oil spillover reaches 49%, with spillover remaining relatively high until

March 2019. From March 2019 - February 2022, gas-to-oil spillover stays below 2% and is followed by elevated spillover at the start of the Russia-Ukraine conflict until February 2023. After this, gas-to-oil spillover stays under 2% except for an 8% spillover value in January 2024 from another winter storm. Together, while some bidirectional spillovers remain, the instances of high oil-to-gas spillover are much less frequent and long periods of low gas-to-oil spillovers return at the end of the sample. This indicates reduced oil-to-gas and gas-to-oil spillovers, with extensions in the length of time with low spillover.

Finally, net spillovers in Figure 5 show neither oil nor gas is the dominant transmitter of volatility, where positive values indicate that gas-to-oil spillovers are stronger than oil-to-gas spillovers (i.e., gas dominance). Except for the peak surrounding 2018-1-2 during the bomb cyclone event, net spillover stays approximately within a ± 10 pp spillover range. Periods of oil dominance appear through 1998, October 2001 – December 2002, most of 2004 - 2007 and also January 2016 – June 2017. The largest negative net spillovers are -8pp in November 2002 and March 2015. Periods of gas dominance include 1999, January - September 2001, 2003, October 2008 - July 2009, December 2011 - September 2013 and July 2017 - December 2018. Moreover, gas dominance appears surrounding the Russia-Ukraine conflict starting approximately in March 2022 and returning to zero in March 2023. Net spillover goes beyond positive 10pp on multiple occasions in December 2002, January 2018 and May 2022. However, neither oil nor gas emerges as the dominant transmitter of volatility because periods of gas or oil dominance are limited to a handful of years.

4.2. Bai-Perron Structural Break Test

From the Diebold-Yilmaz total spillover index, the Bai-Perron test finds four structural breaks on 2002-04-10, 2006-06-28, 2015-01-05 and 2019-03-07 indicated by the vertical lines in Figure 3. Moreover, Table 1 shows the average total and net spillovers for each regime with the main difference being variations in average total spillover. The early sample period is characterized by two regimes of similar length (~ 4 years) from February 1998 - April 2002

and April 2002 - June 2006. Average total spillover in the first regime is 2.45% but increases for the second regime, almost doubling to 4.11%. The main elevation in Regime 2 occurs from late 2002 to early 2004 surrounding conflicts in the Middle East (Sollinger et al., 2008). Regime 3 is the longest regime extending from 2006 - 2015 and is characterized by relatively low average total spillover at 2.16%, consistent with previous studies finding decoupling of volatilities in the post-shale revolution period (Lovcha and Perez-Laborda, 2020). While production from shale fields took off in ~2009, previous studies of the oil-gas cointegrating relationship suggest that the transition between pre-shale and post-shale price dynamics began earlier, around 2005 (Hasanli, 2024), a finding which I support with the period of relatively low spillover beginning in 2006.

Regime 4 begins in January 2015 immediately after an OPEC decision to drive prices down in November 2014 to take market share from higher-cost shale producers (Behar and Ritz, 2016). Regime 4 contains the highest average spillover across any regime, including the global oil supply glut in 2015 where I observe moderate oil-to-gas spillover and Winter Storm Grayson in 2018 with high gas-to-oil spillover. While the structural breaks show periods of relatively high and low total spillover, average total spillover across regimes remains mild to moderate in size. The highest average spillover of 4.16% through 2015-2018 indicates that 4.16% of the total forecast error variance of oil and gas market volatilities can be explained by cross-market spillovers. From past studies (Diebold and Yilmaz, 2012; Lovcha and Perez-Laborda, 2020; Gong et al., 2021), a 4.16% average spillover indicates mild connectedness across markets. However, after 2019, the number dropped to the lowest value among the regimes which indicates a relative decoupling of volatilities, with a 1.69% value indicating very low connectedness.

Table 1 also shows that average net spillovers remained at approximately zero across regimes (ranged from -0.54 to 0.58pp), indicating that neither oil nor gas is dominant across the sample period. While most regimes show positive average net spillover, the magnitudes are not large enough to show dominance of either product in any regime. This is made ap-

parent by the range of net spillover in Figure 5. At its extremes, net spillover goes between $\pm 10\text{pp}$, and in its more moderated times, regularly goes between $\pm 3\text{pp}$. Therefore, even the larger average net spillover in regime three of 0.58pp is not large enough to indicate natural gas dominance. Likewise, the negative value in Regime 2 (-0.54) indicates that spillovers are approximately of equal strength. Because net spillover remains near zero, the reduction in total spillover at the end of the sample is attributable to declines in both oil-to-gas and gas-to-oil spillovers.

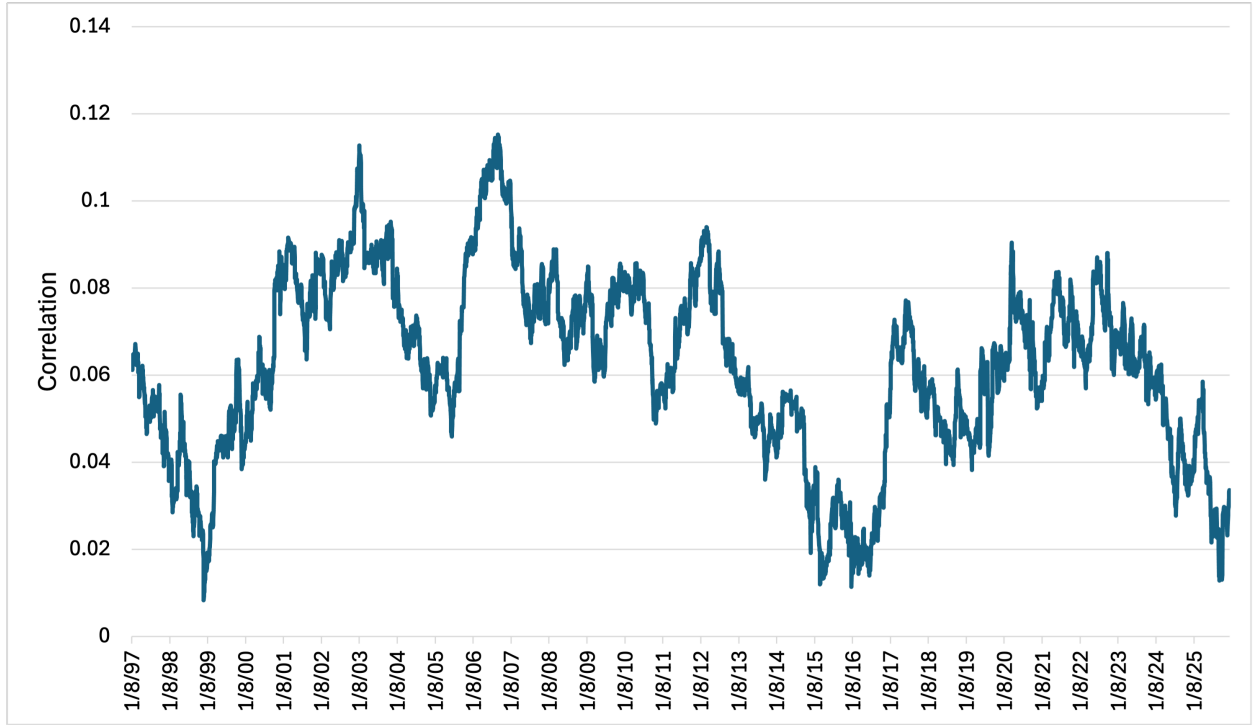


Figure 6: DCC-GARCH dynamic conditional correlation of returns.

4.3 DCC-GARCH

The DCC-GARCH model estimates correlation of returns controlling for time-varying volatility in oil and gas prices. The estimated volatility characteristics are summarized by the coefficients ω , α and β in Table 2 where subscript n indicates natural gas and subscript o indicates oil. Natural gas has a higher unconditional volatility $\omega_n = 0.29$ as opposed to $\omega_o = 0.10$ which is consistent with realized volatilities in Figure 2 where natural gas is more

volatile than oil. Next, $\alpha_n = 0.14$ is larger than $\alpha_o = 0.07$ and $\beta_n = 0.86$ is smaller than $\beta_o = 0.90$, showing that natural gas volatility has greater response to initial price changes, but less persistence between periods compared to oil. Finally, $\alpha_n + \beta_n$ and $\alpha_o + \beta_o$ are both very close to one, indicating slow reversion to unconditional volatility. The properties of the conditional correlation of returns are summarized by parameters a and b in Table 2, where a is near zero and b is very close to one. This means that the correlation of returns is slow to respond to shocks but remains highly stable once shocked. Additionally, $a + b$ is very near one, indicating slow reversion to mean correlation.

Next, Figure 6 shows the estimated conditional correlation ρ_t of oil and gas standardized returns, where the average correlation is approximately 0.06. The conditional correlation shows significant variation across time, with correlation dropping from the sample average to almost to zero in 1998, rising to sample highs of 0.11 in 2003-2006, declining to zero in 2015 and later rising to the range of 0.06 – 0.08 by 2020. The correlation once again begins to decline in 2023, going from 0.07 at the start of the year to zero by the end of the sample. The conditional correlation graph suggests an overall decoupling of the two markets because at the end of the sample returns appear to be less connected than in earlier periods.

5 Discussion

I show daily volatility spillover among US oil and natural gas spot prices has reduced from 1997 – 2025, thereby extending the findings of previous studies which covered a shorter timespan (Lovcha and Perez-Laborda, 2020) or only gave oil-gas spillover results aggregated across time (Kočenda and Moravcová, 2024). I find mild to moderate spillovers across the pre-2019 sample period and a further reduction post-2019. Specifically, total and directional indices show extended periods of low volatility spillover at the end of the sample, and the Bai-Perron structural break test confirms this because the lowest spillover regime occurs after 2019. Regarding magnitudes of previous studies, the 5 – 10% spillovers and the high of 49%

I find before 2019 are consistent with the moderate time-varying connectedness between oil and gas futures found by Lovcha and Perez-Laborda (2020). However, the timing of peaks differs, suggesting differences in spot and futures price spillover. Moreover, the total spillover index is lower than found by Gong et al. (2021), but the discrepancy can be explained by their inclusion of heating oil and gasoline, which plausibly increases connectedness given that the products are derivatives of crude oil. Finally, Kočenda and Moravcová (2024) show low average oil-to-gas and gas-to-oil spillovers from 1997 - February 2023, with the exception of moderate gas-to-oil spillover from February 2022-February 2023 during the Russia-Ukraine war. I likewise find low average spillover across the aggregated windows in Table 1 but do observe periods of moderate to high spillover in disaggregated estimates (Figures 3-4). Moreover, I show that the period of higher spillover during the Russia-Ukraine war dissipates after February 2023, leading to low spillovers through the end of the sample.

The decrease in spillovers after 2019 supports past findings of increased gas market independence (Gong et al., 2021; Lovcha and Perez-Laborda, 2020; Zhu et al., 2018). In particular, Lovcha and Perez-Laborda (2020) show that both oil-to-gas and gas-to-oil spillovers reduced after GFC, which coincided with the shale revolution. Similarly, Gong et al. (2021) show that spillovers among petroleum products and natural gas futures prices reduced after the shale revolution, with the limitation that results are not shown for directional results (oil-to-gas and gas-to-oil spillover indices). Finally, Zhu et al. (2018) confirm these findings using Granger-causality models on implied volatility data from 2007 – 2015, showing little spillover overall, with no gas-to-oil spillover and limited oil-to-gas spillover. Together, these results suggest increased independence of oil and gas market volatilities, which I extend to cover more recent time periods.

The real options theory suggests investment in many natural gas infrastructure projects has been limited by price volatility. Infrastructure investment may be increasing despite the price volatility in recent years because the natural gas price has been rising from improvements in the efficiency of natural gas generators (Hasanli, 2024), external demand from

Europe through LNG exports (Di Bella et al., 2024) and increased power needs from data centers (EIA, 2025). However, transmission of volatility from oil to natural gas could be a serious threatening factor to producers given that any source of volatility brings into question whether their investment will remain profitable. My finding of low volatility spillovers indicates that the firm decision to delay natural gas infrastructure investments is not influenced by oil price volatility. Therefore, models of infrastructure investment decisions can be simplified to a function of natural gas fundamentals rather than oil market volatility. This study extends the finding of volatility independence to a longer time period, which is important because recent changes in oil and gas markets could have shifted the spillover relationship.

Next, I show that neither oil nor natural gas is the dominant transmitter of volatility. The oil-to-gas and gas-to-oil spillover indices tend to be of similar magnitude with periods of positive or negative net spillover limited to a handful of years. This contrasts with past findings of oil-to-gas volatility spillovers being stronger than gas-to-oil spillovers (Gong et al., 2021), which is sensible because oil has historically been the more valuable commodity in the coproduction relationship. Due to this, coproducers made production decisions using oil prices, which led to changes in natural gas supply (Stern, 2020). Moreover, oil price volatility has previously signaled economic instability (Ebrahim et al., 2014) which affects natural gas demand through reduced industrial production. Although having been useful in the past, oil volatility as a macroeconomic driver may be an outdated concept given that use of other energy inputs has increased as a share of overall energy consumption since the 1950's-70's, for instance, when oil price increases signaled future recessions (Hamilton, 1983). Therefore, a possible explanation of reduced oil-to-gas volatility spillover is that energy use has become more dispersed among different products.

My results are consistent with Lovcha and Perez-Laborda (2020) who find alternating spillover dominance and attribute the presence of gas-to-oil spillovers to increased financialization and capacity to profit from existing arbitrages. Kočenda and Moravcová (2024) also

find gas-to-oil spillover surrounding the start of the Russia-Ukraine conflict in 2022. Volatility may have spilled from gas to oil markets at this time because Europe’s main source of energy prior to the conflict was Russian natural gas, and natural gas volatility was signaling that Europeans could not obtain one of their primary sources of energy (Di Bella et al., 2024). This reasonably affects the oil market because actors were uncertain of what damages to European industry would occur from a lack of energy (Di Bella et al., 2024). Similarly, it is possible that volatility in the US gas market both leads consumers to switch among oil and gas for short-term gains and creates general economic uncertainty. The finding that neither oil nor gas is the dominant transmitter is significant because it shows that risk transmission depends on the specific type of shock rather than always coming from the same market.

Finally, DCC-GARCH results show that correlation of returns is decreasing over time. The dynamic correlation among oil and gas returns shows a declining trend at the end of the sample which is consistent with previous findings that oil and gas price levels are increasingly independent of one another (Dahl et al., 2012; Hasanli, 2024; Wang et al., 2025). Early evidence from the oil-gas ratio indicated that if oil rises then natural gas will eventually increase (i.e., natural gas will have a price floor if oil is high – Brown and Yucel 2008), but that appears to be an outdated concept because oil prices were high after the shale revolution and natural gas prices were very low (Hasanli, 2024). This information is useful for producers to signal that diversification among these energy sources is valuable, specifically through shared wells compared to pure natural gas or oil plays because when one price goes down the other does not necessarily also drop. Moreover, correlation of oil and gas returns gives information regarding the expected returns to infrastructure investment. My findings suggest that natural gas price levels, and thus expected returns to investments, are not strongly influenced by oil prices. While previous studies such as Hasanli (2024) and Wang et al. (2025), covering 1997 – 2022 and 2016 – 2023, respectively, extend to a similar timespan as the current study, the DCC-GARCH results show that their findings of weakened correlation of price levels are robust to controlling for time-varying volatility

across the two markets.

Together, the results indicate that oil and gas markets are largely independent of one another, which suggests that oil market shocks are not a significant contributor to the natural gas infrastructure investment decision. In contrast, it appears most natural gas volatility comes from natural gas fundamentals (e.g., weather, storage). Therefore, the significance of reduced volatility spillovers among oil and gas is to eliminate a variable that firms consider in the investment decision, specifically volatility in the oil market. Moreover, the oil market is becoming increasingly uncertain now given widespread geopolitical instability from the Russia-Ukraine and US-Iran wars. Since the US natural gas market is broadly local with most gas consumed in the US being produced domestically (EIA, 2019a), natural gas producers may be willing to invest in domestic natural gas infrastructure despite global conflicts because the natural gas space is not heavily influenced by oil. In conclusion, the current study shows that correlation between oil and natural gas market volatilities has declined from 1997-2025, suggesting that oil and gas markets do not significantly influence uncertainty in the other market.

A limitation of the current study is that the data is restricted to US oil and gas markets, which was chosen because the US is currently among the largest producers and consumers of energy in the world, recently becoming an oil exporter due to the shale revolution (EIA, 2023a). Because natural gas is primarily a regional commodity, it is sensible that the relationship between oil and gas will be different by country, so future studies are needed to examine this. Moreover, future studies could track oil volatility spillovers into more local natural gas spot markets within the US. Furthermore, I limit my analysis to spot instead of futures markets as a gauge on fundamental factors related to these commodities as opposed to more complex dynamics affecting futures related to interest rates and trading of paper contracts whose intent is never to be called for delivery (Cheng and Xiong, 2014; Pindyck, 2001). Therefore, future studies should compare volatility spillover results using spot and futures contracts. It is also possible that the way I estimated volatility could affect the final

results given differences in the persistence of volatility between natural gas and crude oil. For instance, when I calculate the standard deviation of returns across 20 days instead of a shorter time frame, it is plausible that relative volatility across oil and gas changes and influences these results. To overcome this limitation, I estimated 10- and 30-day window width realized volatilities as a robustness check (Appendix Figures A1-6 and Tables A1-2) indicating the magnitude of spillover remains fairly stable across all these conditions.

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A ADDITIONAL RESULTS

A.1 Results using 10-day and 30-day rolling window realized volatility

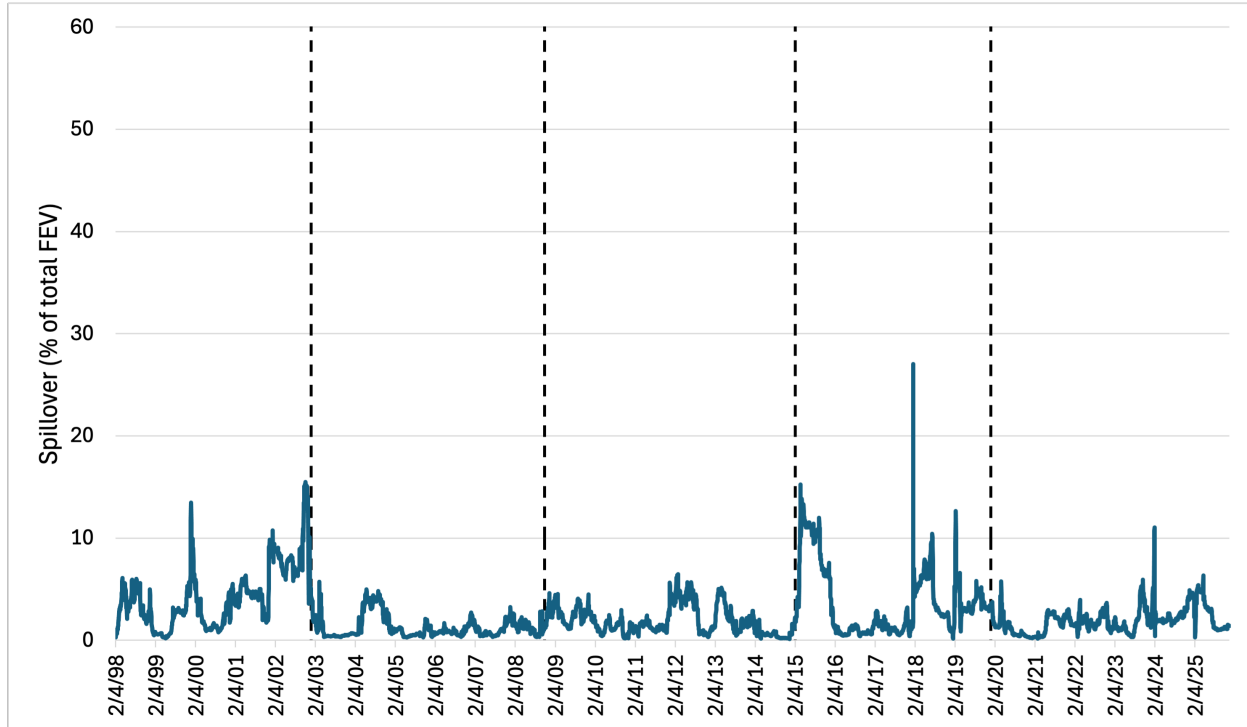


Figure A.1: Diebold-Yilmaz total volatility spillover index on 10-day rolling window realized volatilities. Vertical lines indicate structural breaks found by Bai-Perron test on 2002-12-26, 2008-10-30, 2015-02-03 and 2019-12-26.

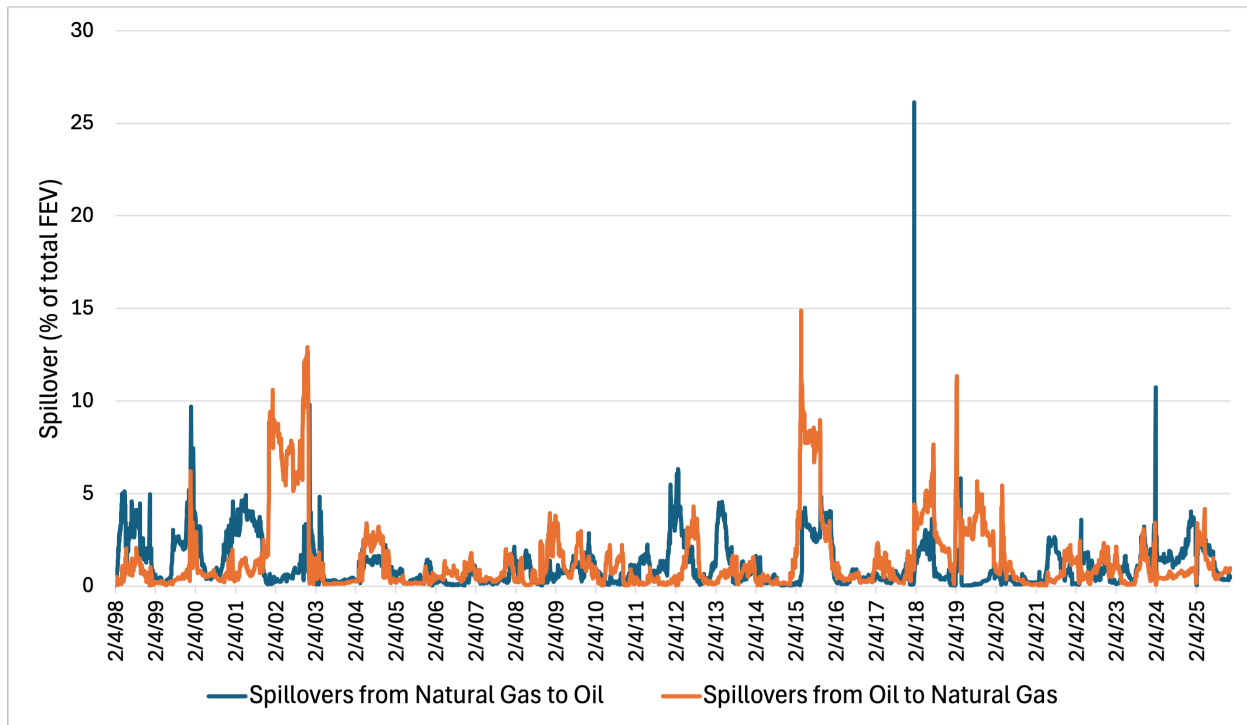


Figure A.2: Diebold-Yilmaz directional volatility spillover indices on 10-day rolling window realized volatilities.

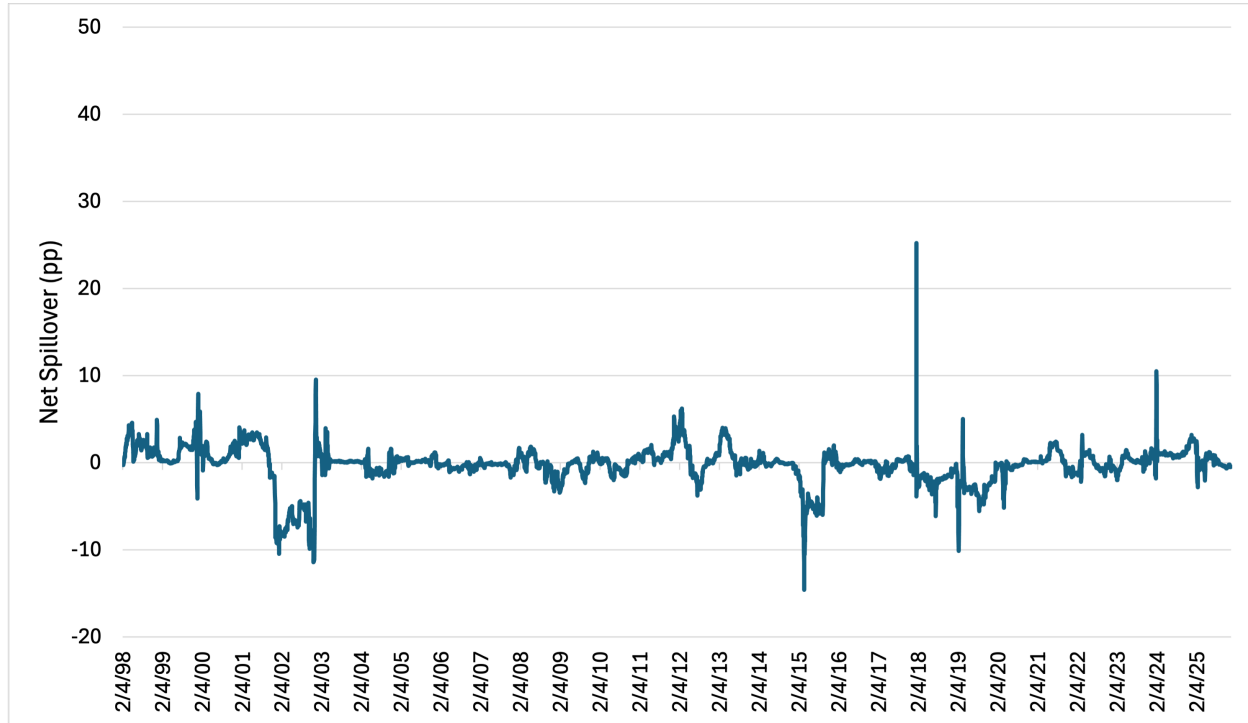


Figure A.3: Diebold-Yilmaz net volatility spillover index on 10-day rolling window realized volatilities. Positive values indicate gas-to-oil spillover is stronger than oil-to-gas spillover.

Table A.1: Summary of volatility spillover results on 10-day rolling window realized volatilities within regimes identified by Bai-Perron test.

Regime	Start Date	End Date	Duration	Average Total Spillover (%)	Average Net Spillover (pp)
1	Jan 1998	Dec 2002	4.9 years	4.19	−0.24
2	Dec 2002	Oct 2008	5.8 years	1.26	−0.10
3	Oct 2008	Feb 2015	6.3 years	1.89	0.17
4	Feb 2015	Dec 2019	4.9 years	3.77	−1.62
5	Dec 2019	Dec 2025	6.0 years	1.92	0.23

Table A.2: Summary of volatility spillover results on 30-day rolling window realized volatilities within regimes identified by Bai-Perron test.

Regime	Start Date	End Date	Duration	Average Total Spillover (%)	Average Net Spillover (pp)
1	Feb 1998	Jan 2003	4.9 years	2.80	−0.36
2	Jan 2003	Jan 2012	9.0 years	1.41	0.25
3	Jan 2012	Oct 2021	9.7 years	1.91	0.22
4	Oct 2021	Dec 2025	4.2 years	3.14	−0.91

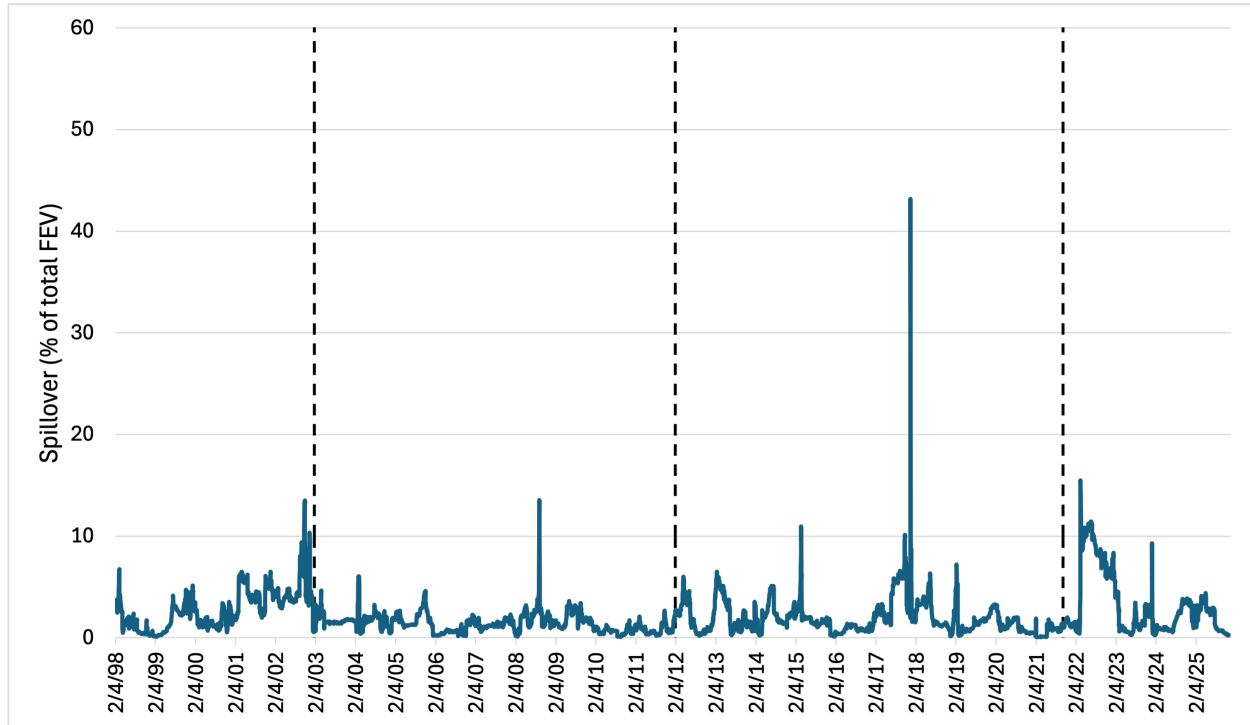


Figure A.4: Diebold-Yilmaz total volatility spillover index on 30-day rolling window realized volatilities. Vertical lines indicate structural breaks found by Bai-Perron test on 2003-01-24, 2012-01-30 and 2021-10-05.

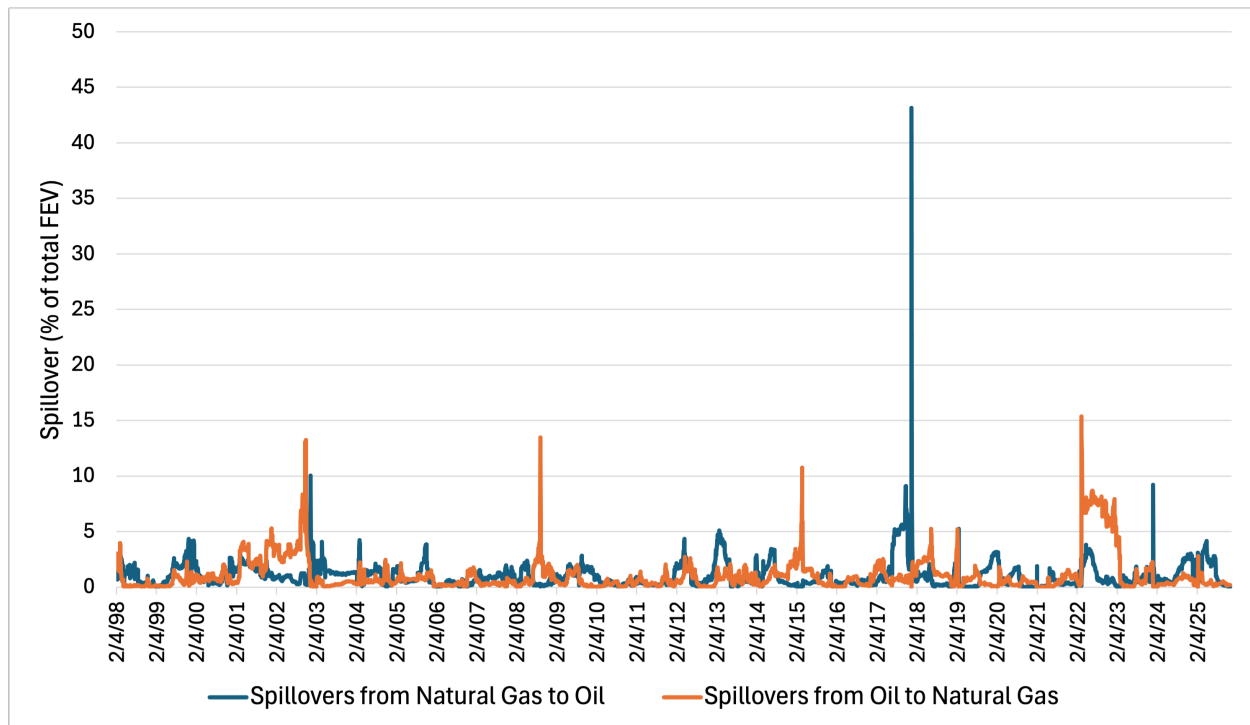


Figure A.5: Diebold-Yilmaz directional volatility spillover indices on 30-day rolling window realized volatilities.

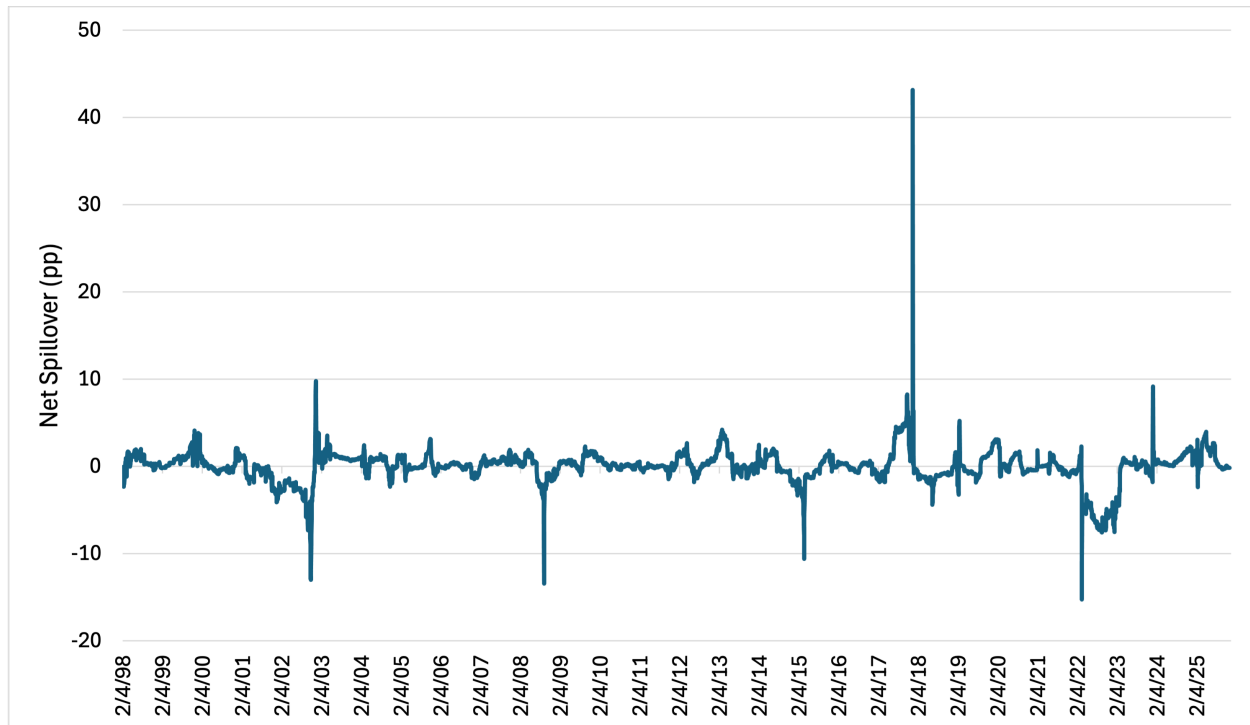


Figure A.6: Diebold-Yilmaz net volatility spillover index on 30-day rolling window realized volatilities. Positive values indicate gas-to-oil spillover is stronger than oil-to-gas spillover.

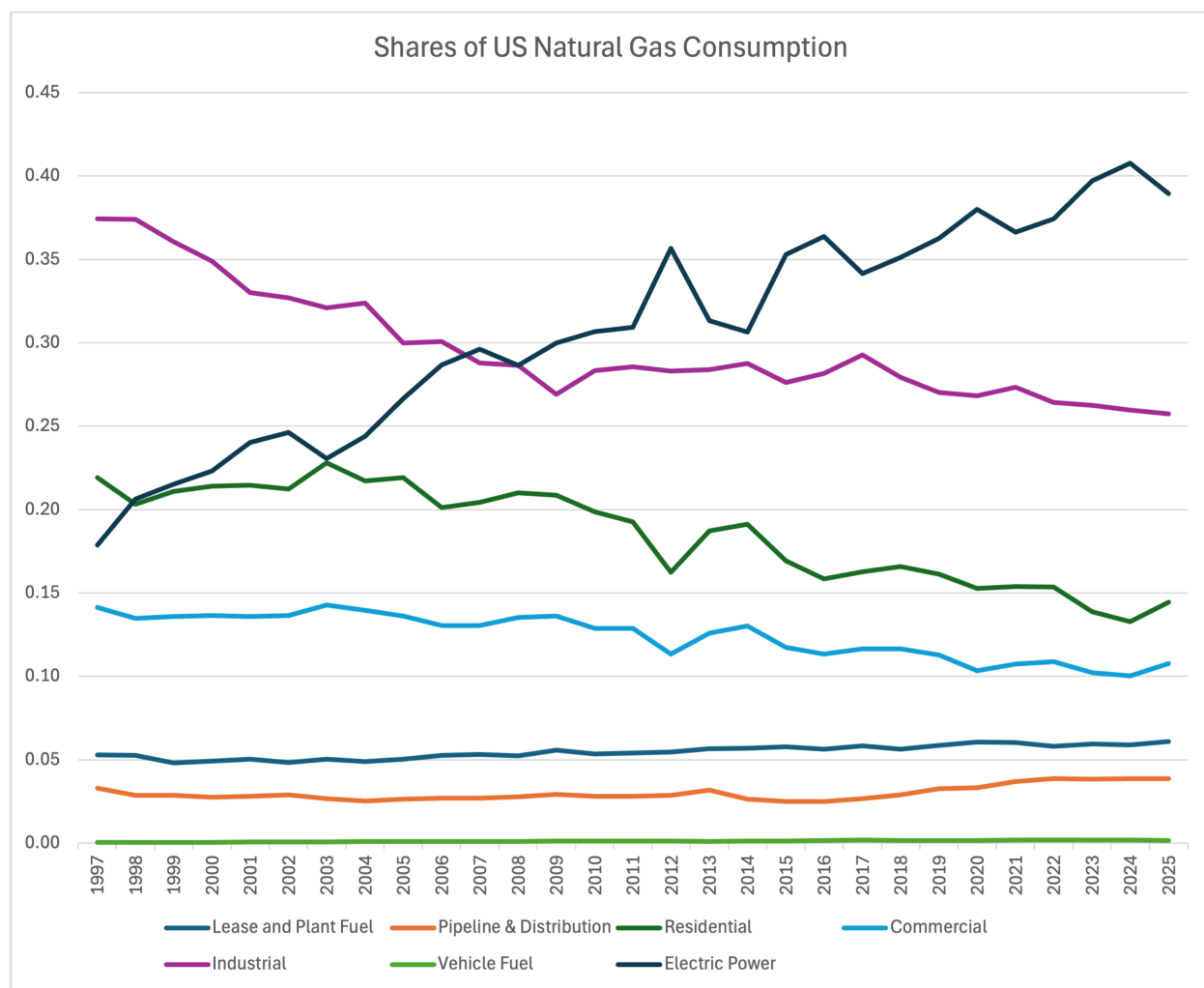


Figure A.7: US natural gas consumption shares from EIA data at this link